

WHATISAPEPTIDE.AI

The Peptide Playbook

An evidence-graded guide to what peptides are, what they actually do, and how to tell the difference.

Every compound rated by the strength of its human evidence — Approved, Clinical, Preclinical or Speculative — with sources cited, current regulatory status through 2026, and a frank account of what remains unknown.

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For educational and informational purposes only. Not medical advice. Many compounds described in this guide are not approved for human use by the FDA or any comparable regulator. This guide does not provide dosing for unapproved compounds. See the full disclaimer at the back.

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Start Here

Peptides have become one of the most talked-about topics in health — and one of the most confusing.

Walk into the conversation cold and you'll find the same word doing an enormous amount of work. "Peptide" describes insulin, which has saved lives for a century. It describes semaglutide, a rigorously tested medicine that reshaped obesity treatment. And it describes compounds sold online in vials labelled "research use only," whose entire human evidence base could be counted on two hands.

Those things are not alike. Treating them as if they were is how people get hurt, get scammed, or get disappointed.

This guide exists to draw the lines clearly. For every compound covered here, we'll tell you what it is, what it's supposed to do, what has actually been demonstrated in humans, what its legal status is right now, and what remains unknown. Where the evidence is strong, we'll say so plainly. Where it's thin, we'll say that just as plainly — including for compounds that are wildly popular.

You'll notice this guide says "we don't know" more often than most peptide content does. That isn't hedging. It's the most useful thing we can tell you, because the gap between what a compound is *claimed* to do and what has been *shown* is where nearly all the risk in this space lives.

What this guide will not do

Being upfront about this saves everyone time.

It will not tell you what to take. That's a decision for you and a licensed clinician who knows your medical history, your medications, and your goals. No book can do that job.

It will not give you injection protocols for unapproved compounds. You'll find this is where we differ most from other peptide guides. For FDA-approved medicines, we list the approved label dose — because that figure was established through clinical trials and exists in a public regulatory document you can verify. For everything else, there is no authoritative human dose, because the studies that would establish one have not been done. Numbers circulating online for compounds like BPC-157 or ipamorelin did not come from dose-finding trials. They came from forums, vendors, and extrapolation from animal research. Printing them would dress up guesswork as guidance.

It will not sell you anything. There are no product recommendations in these pages and no affiliate links. If a compound has weak evidence, we say so even when it's a bestseller.

How to read the evidence ratings

Every compound in this guide carries one of four ratings. They describe *how much is known*, not how promising something sounds.

APPROVED — Cleared by the FDA for a specific medical use after large human trials. Available by prescription. This is the highest tier of evidence that exists, and only a handful of compounds in this guide reach it.

CLINICAL — Real human trials have been conducted and published. The compound may be investigational, approved only outside the United States, or approved for a different use than the one it's popular for. Evidence exists and can be examined.

PRECLINICAL — Evidence comes from cell cultures and animal studies. The biology may be genuinely interesting. But animals are not people, and the history of medicine is substantially a history of compounds that worked beautifully in mice and did nothing in humans. Most popular "wellness peptides" sit here.

SPECULATIVE — Mechanism and theory, with minimal or no published data even in animals for the use being promoted. Treat marketing claims at this tier as claims, not findings.

A low rating doesn't mean a compound is worthless. Some of tomorrow's important medicines are sitting at PRE-CLINICAL today. It means the certainty isn't there yet, and you should size your expectations — and your risk tolerance — accordingly.

Part I — Peptides, Demystified

What a peptide actually is

A peptide is a short chain of amino acids. Amino acids are the same building blocks that make up protein in your food and in your body. String a few dozen together and you have a peptide; string hundreds together and fold it up and you have a protein. The line between the two is a matter of convention rather than chemistry — somewhere around 50 amino acids is the usual cutoff.

Your body makes thousands of them, and uses them as signals. Insulin is a peptide. So is glucagon. So are the hormones that govern hunger, satiety, growth, stress, and reproduction. When your gut detects a meal, it releases peptide messengers that travel to your brain and pancreas carrying specific instructions: release insulin, slow digestion, stop eating.

The therapeutic idea follows directly. If the body already runs on these messages, you can intervene by supplying a message of your own — a synthetic peptide that mimics, amplifies, or blocks a natural signal. Because each peptide is shaped to fit one particular receptor, the intervention can be far more targeted than an older small-molecule drug that interacts with many systems at once.

That specificity is the genuine appeal of peptide medicine. It is also why peptides are not magic: a precisely targeted signal still only does what that one receptor does.

Why they're usually injected

Peptides have a practical problem. Your digestive system is built to break proteins down into amino acids, and it does this to peptides too. Swallow most peptides and they are destroyed before reaching the bloodstream.

That's why the great majority are given by injection under the skin, which bypasses the gut entirely. Some are delivered as nasal sprays, which offer a route to the bloodstream across the thin nasal membrane. A few are formulated as creams for local effect on skin.

Oral peptide drugs do exist, but they require serious engineering. Oral semaglutide is co-formulated with an absorption enhancer and must be taken on an empty stomach with strict timing rules, and even then only a small fraction is absorbed. When a supplement company claims their oral peptide capsule delivers the same compound as an injection, that claim deserves hard scrutiny — the chemistry is genuinely difficult and the burden of proof sits with the seller.

A related point worth internalising: peptides also break down quickly *in* the body. Natural GLP-1 has a half-life of roughly two minutes. The central achievement behind the modern weight-loss drugs wasn't discovering GLP-1 — it was re-engineering it so that enzymes couldn't destroy it and it could bind to a carrier protein in blood, stretching those two minutes into about a week. That engineering is what made a once-weekly injection possible, and it represents decades of pharmaceutical chemistry, not a simple copy of a natural hormone.

Why the conversation exploded

Two things happened close together.

First, the engineering problem above was solved, making convenient dosing realistic. Second, the GLP-1 drugs produced weight-loss results that no previous medication had approached — results that, in some trials, edge into territory once reserved for bariatric surgery. That combination generated enormous public attention and enormous commercial interest.

The commercial interest is the part to watch. The success of a handful of rigorously tested peptide drugs created a halo that now extends over hundreds of compounds that share only a chemical category with them. A vendor selling an untested peptide benefits from the credibility earned by semaglutide, without having done any of the work that earned it.

Recognising that borrowed credibility is most of what it takes to navigate this space intelligently.

Part II — Where the Law Actually Stands

This chapter matters more than most readers expect, because a compound's legal status tells you a great deal about how thoroughly it has been examined — and determines whether what you can buy has been made under any meaningful quality oversight.

The three categories that matter

FDA-approved drugs. The manufacturer ran clinical trials, submitted the data, and the FDA concluded the benefits outweigh the risks for a specific population and use. The product is made under pharmaceutical manufacturing standards, the label is legally binding, and side effects are tracked after launch. Semaglutide, tirzepatide, tesamorelin, teriparatide, abaloparatide, bremelanotide and afamelanotide are in this group.

Compounded preparations. Compounding pharmacies prepare customised medications for individual patients under Sections 503A and 503B of the Federal Food, Drug, and Cosmetic Act. Compounded drugs are not FDA-approved and do not undergo premarket review for safety or efficacy. For a pharmacy to compound with a given bulk substance under 503A, that substance must have a USP or National Formulary monograph, be a component of an approved drug, or appear on the FDA's 503A Bulks List.

"Research use only" products. Sold online, labelled not for human consumption, and outside the prescription drug supply chain entirely. There is no regulatory review of identity, purity, sterility, or dose. The "research use only" label is the legal mechanism that allows the sale; it is not a description of who buys the product.

What changed in 2023, and again in 2026

This area has moved considerably, and much of what you'll read online is out of date.

In September 2023, the FDA placed a large group of popular peptides into **Category 2** of the interim 503A Bulks List — the designation for substances that raise significant safety concerns [20]. The substances added included AOD-9604, BPC-157, LL-37, CJC-1295, dihexa acetate, emideltide (DSIP), epitalon, injectable GHK-Cu, ipamorelin acetate, kisspeptin-10, KPV, Melanotan II, PEG-MGF, MOTS-c, selank, semax, thymosin alpha-1 and a thymosin beta-4 fragment [20]. The practical effect was that licensed compounding pharmacies could no longer prepare them, while the unregulated online market was untouched.

Then the picture shifted. On April 15, 2026, the FDA removed BPC-157, along with eleven other peptides, from Category 2 entirely [21]. That sounds like good news, and it is widely misread as such. But the FDA did not move them to Category 1 — the list of substances permitted for compounding — and neither BPC-157 nor TB-500 is FDA-approved or has a recognised USP/NF monograph, leaving them in a regulatory gray zone where they are neither explicitly prohibited nor authorised [21].

Most recently, the FDA's Pharmacy Compounding Advisory Committee met on July 23–24, 2026 to consider seven peptides for the 503A Bulks List. The committee voted in favour of BPC-157, KPV, TB-500, MOTS-c, epitalon and semax, while recommending against emideltide [22]. Two details make this vote less decisive than the headlines suggested. FDA staff had recommended against inclusion of all the peptides before the committee

reached a different conclusion, and the votes did not make these peptides lawful for compounding — the PCAC serves an advisory role, and the FDA retains authority to decide what is placed on the list [22]. All six favourable votes passed by narrow margins [23].

What this means for you

Three things, and they hold regardless of how the rulemaking eventually lands.

An advisory vote is not an approval. None of these compounds became an FDA-approved medicine in 2026. Inclusion on a compounding list — if it happens — would mean a pharmacy may legally prepare the substance on a prescription. It would not mean the compound has been shown to work.

Regulatory movement does not create evidence. The reason the FDA's own reviewers recommended against these substances is that the underlying human studies remain short in duration, small in sample size, and insufficient to establish safety or efficacy for the proposed indications [23]. A committee vote did not change that.

Nothing here legitimises the online research-chemical market. Whatever happens with compounding rules applies to licensed pharmacies dispensing against a prescription. It has no bearing on products sold in vials from unverified suppliers.

Because this area is actively moving, treat any regulatory statement in this guide — including these — as accurate at the time of writing and worth re-checking before you act on it.

A note on sport

Several compounds in this guide are prohibited in competitive sport. WADA's S0 category covers substances with no approval from any governmental health authority for human therapeutic use, which sweeps in most research peptides by definition. Growth-hormone-releasing peptides and secretagogues are prohibited under S2. If you compete in a tested sport, assume a research peptide is banned unless you have confirmed otherwise with your anti-doping authority.

Part III — The Weight-Loss Class

This is where the evidence is strongest, by a wide margin. It is also the only category in this guide where multiple compounds have been tested in tens of thousands of people and demonstrated benefits on outcomes that matter, not just on biomarkers.

These drugs mimic incretins — gut hormones released after eating that signal fullness, slow stomach emptying and improve insulin response. Understanding them well is worth the effort, because they set the standard against which everything else in this book should be measured.

Semaglutide

APPROVED. Sold as Ozempic (type 2 diabetes), Wegovy (weight management) and Rybelsus (oral, diabetes), from Novo Nordisk. A GLP-1 receptor agonist.

What it does. Semaglutide binds the GLP-1 receptor, reducing appetite and food intake, slowing gastric emptying and improving glucose-dependent insulin secretion. Structural modifications protect it from enzymatic breakdown and let it bind albumin in the blood, producing a half-life long enough for weekly dosing.

What the trials showed. In STEP 1, a 68-week randomised trial of 1,961 adults with overweight or obesity without diabetes, once-weekly semaglutide 2.4 mg produced a mean weight reduction of 14.9% compared with 2.4% on placebo [1]. The cardiovascular question was answered separately by SELECT, which enrolled over 17,000 adults with cardiovascular disease and overweight or obesity but without diabetes: semaglutide reduced major adverse cardiovascular events by about 20% relative to placebo over a mean of roughly 40 months [2]. That second result matters enormously — it moved semaglutide from a drug that changes a number on a scale to one shown to prevent heart attacks and strokes.

An oral formulation for weight management was subsequently approved in December 2025.

Approved dosing. For weight management, escalated over approximately 16 weeks from 0.25 mg to 2.4 mg once weekly. Prescription only, under medical supervision.

Known harms. Gastrointestinal effects dominate — nausea, vomiting, diarrhoea, constipation — typically worst during escalation. Less common but serious risks include pancreatitis and gallbladder disease. It carries a boxed warning regarding thyroid C-cell tumours observed in rodents and is contraindicated in people with a personal or family history of medullary thyroid carcinoma or MEN 2. Weight is substantially regained after discontinuation, which is why it's framed as long-term therapy rather than a course of treatment.

Tirzepatide

APPROVED. Sold as Mounjaro (diabetes) and Zepbound (weight management, obstructive sleep apnoea) from Eli Lilly. A dual GIP and GLP-1 receptor agonist.

What it does. Tirzepatide activates two incretin receptors rather than one. The contribution of the GIP arm is still debated mechanistically, but the clinical effect is consistently larger than GLP-1 alone.

What the trials showed. In SURMOUNT-1, 2,539 adults with obesity or overweight received tirzepatide or placebo for 72 weeks. Mean weight reduction reached approximately 21% at the higher doses, with placebo at about 3% [3]. In head-to-head comparison against semaglutide, tirzepatide produced greater weight reduction. It is also approved for moderate-to-severe obstructive sleep apnoea in adults with obesity.

Approved dosing. Started at 2.5 mg weekly and escalated in 2.5 mg steps at four-week intervals to a maintenance dose of 5, 10 or 15 mg once weekly. Prescription only.

Known harms. Essentially the same profile as semaglutide: predominantly gastrointestinal, worst during escalation, with the same boxed warning for thyroid C-cell tumours and the same contraindications. Weight regain after stopping is likewise expected.

Retatrutide

CLINICAL — investigational, not approved anywhere. A once-weekly triple agonist at the GIP, GLP-1 and glucagon receptors, from Eli Lilly.

What it does. Retatrutide adds glucagon receptor activity to the dual-agonist template. Glucagon receptor agonism is thought to increase energy expenditure and reduce hepatic fat, complementing the appetite suppression of the incretin arms.

What the trials showed. The Phase 2 trial published in 2023 reported mean weight reduction of 24.2% at 48 weeks on the highest dose, with the weight curve not yet plateauing [4]. Phase 3 results have since arrived. In TRIUMPH-4, in adults with obesity and knee osteoarthritis, retatrutide reduced weight by up to an average of 28.7% and reduced pain by up to 75.8% [27]. In TRIUMPH-1, reported in May 2026 across 2,339 participants, mean weight reduction at 80 weeks was 17.6% on 4 mg, 23.7% on 9 mg and 25.0% on 12 mg, against 3.9% on placebo; a subgroup escalated to maximum tolerated dose reached up to 30% reduction at 104 weeks [28].

Dosing. No approved dose exists. Trials escalated to maintenance doses of 4, 9 or 12 mg weekly.

Known harms. Gastrointestinal effects at higher rates than the dual agonists, along with increases in heart rate. Because glucagon receptor agonism affects hepatic glucose output, glycaemic and liver parameters were monitored closely in trials.

The honest frame. These are the largest weight-loss figures ever reported in Phase 3 obesity trials. They are also results from a compound that no regulator has yet reviewed and approved. Material sold online as "retatrutide" has no connection to Lilly's manufacturing, and none of the safety monitoring that accompanied these trials applies to it.

CagriSema

CLINICAL — investigational, under FDA review. A fixed-dose combination of cagrilintide, a long-acting amylin analogue, with semaglutide, from Novo Nordisk.

What it does. Amylin is a satiety hormone co-secreted with insulin that signals fullness through a pathway distinct from GLP-1. Combining the two targets appetite from two directions.

What the trials showed. In REDEFINE 1, participants who adhered to treatment lost an average of 22.7% of body weight over 68 weeks [29]. In adults with type 2 diabetes (REDEFINE 2), the reduction was smaller, as is typical for this class in diabetic populations. Worth knowing for balance: a head-to-head trial against tirzepatide reported in 2026 did not demonstrate non-inferiority, which tempered expectations that the combination would displace the leading dual agonist.

Novo Nordisk filed a New Drug Application with the FDA for CagriSema in adults with obesity, or overweight with at least one weight-related comorbidity [30]. As of early 2026 it was not FDA approved, with review expected during 2026 [30].

Dosing. Cagrilintide 2.4 mg with semaglutide 2.4 mg, once weekly, in trials. No approved dose.

Known harms. Gastrointestinal effects consistent with the incretin class, plus injection-site reactions.

Survodutide and mazdutide

CLINICAL. Two further dual agonists worth knowing by name.

Survodutide (Boehringer Ingelheim) is a GLP-1 and glucagon receptor dual agonist. Phase 3 data reported in 2026 showed mean weight reduction of roughly 16.6% at 76 weeks, and Phase 2 results in metabolic dysfunction-associated steatohepatitis were encouraging. Not FDA approved.

Mazdutide (Innovent) is also a GLP-1 and glucagon dual agonist, and became the first drug of its class approved for weight management anywhere when it was cleared in China in 2025. Trial data showed weight reductions in the mid-teens as a percentage, with notable reductions in liver fat. It is not FDA approved, and most of its evidence base comes from Chinese populations.

Orforglipron — and why it belongs in a peptide guide

APPROVED. Approved by the FDA on April 1, 2026 as Foundayo, the first oral non-peptide GLP-1 receptor agonist; in the ATTAIN-1 Phase 3 trial it produced 12.4% weight loss at 72 weeks [31].

It's included here for an instructive reason: **orforglipron is not a peptide.** It is a small molecule that happens to activate the same receptor. Because it isn't a peptide, it survives digestion and can be taken as a daily pill without the timing restrictions oral semaglutide requires.

The lesson generalises. What matters therapeutically is the receptor and the evidence, not whether a molecule falls into the chemical category "peptide." The category is not itself a virtue, and marketing that treats "it's a peptide" as a selling point is trading on a halo rather than an argument.

AOD-9604 and HGH Fragment 176-191

PRECLINICAL for the promoted use, with a negative human trial. These are fragments of the growth hormone molecule marketed for fat loss.

The rationale was appealing: isolate the lipolytic portion of growth hormone while leaving behind its effects on growth and glucose metabolism. It is a genuinely elegant idea. It was tested, and in a large Phase 2b obesity trial

AOD-9604 did not separate from placebo on weight loss. Development for obesity was discontinued; later work explored cartilage and joint applications.

HGH Fragment 176-191 is closely related and is frequently marketed using AOD-9604's mechanism story — while quietly omitting that the mechanism story ends in a failed trial.

This is one of the clearest cases in the guide where an attractive hypothesis was properly tested and did not hold up. Marketing in this category continues largely unchanged.

Part IV — Recovery and Repair

This is the most popular category in consumer peptide use and the one with the widest gap between enthusiasm and evidence. The mechanisms are real and the animal literature is often striking. Human trials are largely absent.

That combination — compelling biology, minimal human data, heavy marketing — is precisely the pattern that should slow you down.

BPC-157

PRECLINICAL. A synthetic 15-amino-acid peptide derived from a sequence identified in gastric juice. Marketed extensively for tendon, ligament, muscle and gut healing.

What the animal research shows. Rodent studies report accelerated tendon and ligament healing, tendon-to-bone repair, protection of the gut lining, and promotion of angiogenesis — new blood vessel formation into injured tissue. The body of work is large and consistent, and the proposed mechanisms are coherent.

What the human research shows. Very little. As of 2025, only a small number of pilot studies totalling roughly 30 human subjects had been published. BPC-157 has not been approved for use in standard medicine by the FDA or other global regulatory authorities due to the absence of sufficient and comprehensive clinical studies confirming its health benefits in humans [17]. That evidence gap was central to the FDA's original safety concerns, which focused on immunogenicity and impurities.

Dosing. There is no established human dose. No dose-ranging trial in humans has been published. The figures circulating in vendor literature and forums are not derived from clinical research.

Status. Removed from Category 2 in April 2026 without being added to Category 1; recommended for the 503A Bulks List by the PCAC in July 2026, pending FDA decision. Not FDA-approved for any use. It is prohibited by the World Anti-Doping Agency under its S0 category for substances with no approval from any health authority [21], [26].

The honest frame. BPC-157 is the most popular compound in this guide and among the least studied in humans. Those two facts sit together uncomfortably and are worth holding in mind. The animal data justify serious clinical investigation. They do not substitute for it.

TB-500 and thymosin beta-4

PRECLINICAL. A distinction worth getting right: thymosin beta-4 is a naturally occurring 43-amino-acid protein involved in wound repair. "TB-500" generally refers to a shorter synthetic fragment. They are frequently discussed as though interchangeable. They are not the same molecule, and product identity in the consumer market varies.

What the research shows. Thymosin beta-4 promotes cell migration into damaged tissue, supports angiogenesis and modulates inflammation and scarring. The strongest human data belong to full-length thymosin beta-4, prin-

cipally in ophthalmic wound healing, where it has been studied in controlled trials. For the fragment sold in recovery circles, evidence is preclinical.

Dosing. No established human dose.

Status. Not FDA-approved. Recommended for the 503A Bulks List by the PCAC in July 2026 [22], pending FDA decision. Prohibited in sport.

GHK-Cu

CLINICAL for topical cosmetic use; PRECLINICAL for injection. A naturally occurring copper-binding tripeptide. This is the best-evidenced compound in this chapter, provided you're precise about which use you mean.

What the research shows. GHK-Cu stimulates collagen and elastin synthesis, supports antioxidant systems and participates in tissue remodelling. Plasma levels decline substantially with age. Controlled human studies of topical GHK-Cu formulations have reported improvements in skin density, fine lines and collagen content, and the peptide is widely used as a cosmetic ingredient [6].

The important distinction. That human evidence concerns creams and serums applied to skin. It does not transfer to injected GHK-Cu, which has a different exposure profile and a much thinner evidence base. The FDA has treated the two routes differently: GHK-Cu was added to the Category 1 list, but not for injectable routes of administration [21], with injectable GHK-Cu placed in Category 2 in 2023.

Dosing. Topical formulations in cosmetic use typically fall in the range of roughly 0.05–3% concentration. No established dose for injection.

KPV

PRECLINICAL. A tripeptide fragment of alpha-melanocyte-stimulating hormone, retaining anti-inflammatory activity without the pigmentary effects of the parent molecule.

What the research shows. KPV suppresses NF-κB signalling, a central inflammatory pathway, and can be taken up directly by intestinal epithelial cells. Animal models of colitis show reduced inflammation. The mechanistic work is clear and the preclinical data are reasonably consistent.

What's missing. Human trials for the promoted uses.

Status. Not FDA-approved. Recommended for the 503A Bulks List by the PCAC in July 2026 [22], pending FDA decision. No established human dose.

ARA-290 (cibinetide)

CLINICAL. A peptide derived from erythropoietin that retains EPO's tissue-protective signalling while lacking its haematopoietic activity — meaning it does not raise red cell mass or thicken blood.

What the research shows. This one has a genuine, if small, human trial. A randomised controlled study in patients with sarcoidosis-associated small-fibre neuropathy reported improvements in pain and function, with no in-

crease in blood pressure or thrombotic markers. The mechanism — separating EPO's protective effects from its blood-building effects — is well characterised.

The honest frame. Encouraging early human data in a specific neuropathic pain population. Not approved, and the trial was small. This is a compound where continued clinical development is genuinely warranted.

LL-37

PRECLINICAL, with a genuine two-sided risk profile. A human cathelicidin antimicrobial peptide, part of innate immune defence. It kills bacteria, disrupts biofilms and contributes to wound repair.

The nuance that marketing tends to omit: LL-37 is implicated in the pathology of inflammatory skin disease. Elevated cathelicidin activity is a recognised feature of rosacea, and LL-37 has been implicated in psoriasis pathogenesis. This is not a peptide where more is straightforwardly better — the same molecule that supports healing at physiological levels drives inflammation in excess.

Not FDA-approved. No established human dose.

Part V — Growth Hormone Secretagogues

These compounds prompt the pituitary to release its own growth hormone, rather than supplying growth hormone directly. Two natural pathways drive GH release — the GHRH pathway and the ghrelin pathway — and somatostatin restrains it. Most compounds here act on one of the two accelerators.

A framing point before the individual entries. Growth hormone is not a general-purpose vitality molecule. In adults with genuine deficiency, replacement produces real benefits. In adults with normal GH, the effects of raising it are considerably less impressive than marketing suggests, and the metabolic consequences — reduced insulin sensitivity, fluid retention, joint pain — are consistent and dose-related. Raising IGF-1 is not a free action.

Tesamorelin

APPROVED. Sold as Egrifta. A GHRH analogue, and the only FDA-approved GH-releasing peptide in this guide.

What the trials showed. Approved for reduction of excess visceral abdominal fat in people with HIV-associated lipodystrophy. In the pivotal Phase 3 programme, tesamorelin significantly reduced visceral adipose tissue relative to placebo, with associated improvements in triglycerides [7]. A later randomised trial demonstrated reductions in liver fat and prevention of hepatic fat accumulation in the same population [8].

Approved dosing. 2 mg by subcutaneous injection once daily. Prescription only.

Known harms. Arthralgia, peripheral oedema, injection-site reactions, and reduced insulin sensitivity — glucose is monitored during treatment. The visceral fat reduction reverses on discontinuation.

Note on scope. The approval is specific to HIV-associated lipodystrophy. Use outside that population is off-label and not supported by the approval evidence.

Sermorelin

Historically APPROVED; withdrawn. A GHRH fragment, formerly marketed as Geref for paediatric growth hormone deficiency and as a diagnostic agent. It was withdrawn from the US market for commercial rather than safety reasons and now appears mainly through compounding pharmacies.

Its appeal is that it acts upstream, prompting endogenous GH release with negative feedback loops intact, which in principle limits supraphysiological exposure. Evidence for the anti-aging and body composition uses it is now marketed for is limited. Prohibited in sport.

CJC-1295 and ipamorelin

PRECLINICAL to early CLINICAL. These are usually discussed together because they are the most commonly combined pair.

CJC-1295 is a modified GHRH analogue. The version with DAC (drug affinity complex) binds albumin, extending activity to several days. Early human pharmacology studies in healthy adults demonstrated sustained elevation of GH and IGF-1 following single doses, establishing that it does what it's designed to do biochemically. Whether that translates into meaningful clinical benefit has not been established in controlled trials.

Ipamorelin is a selective ghrelin receptor agonist. Its distinguishing feature, demonstrated in the foundational pharmacology work, is that it stimulates GH release without meaningful effects on cortisol, prolactin or appetite — unlike earlier GHRPs. That selectivity is real and is why it became the preferred partner in combinations.

What's missing for both. Controlled trials examining clinical outcomes — body composition, recovery, function — rather than hormone levels. Elevated IGF-1 is a biomarker, not a result.

Status. Neither is FDA-approved. Both were placed in Category 2 in 2023; CJC-1295 was among those removed from Category 2 in April 2026 without being moved to Category 1. No established human doses. Prohibited in sport.

Known concerns. The predictable consequences of sustained GH elevation: fluid retention, arthralgia, carpal tunnel symptoms and reduced insulin sensitivity. People with impaired glucose tolerance should take that last point seriously.

GHRP-2, GHRP-6 and hexarelin

CLINICAL to PRECLINICAL. The earlier generation of ghrelin-pathway secretagogues, all of which release GH effectively but with less selectivity than ipamorelin.

GHRP-2 (pralmorelin) is approved as a diagnostic agent in Japan. GHRP-6 is notable for pronounced appetite stimulation, a consequence of ghrelin receptor activity. Hexarelin is potent but shows tachyphylaxis — diminishing response with repeated dosing — and has additional cardiac tissue activity.

All three can elevate cortisol and prolactin to varying degrees. None is FDA-approved. All are prohibited in sport.

IGF-1 LR3 and IGF-1 DES — a caution

PRECLINICAL, with meaningful theoretical risk. These are modified forms of insulin-like growth factor-1 engineered to evade the binding proteins that normally regulate IGF-1 activity, substantially extending their action.

Two concerns deserve explicit statement. First, IGF-1 has insulin-like activity and can cause hypoglycaemia. Second, IGF-1 signalling is mitogenic — it drives cell proliferation broadly — and elevated IGF-1 levels are associated in epidemiological research with increased risk of several cancers. Deliberately producing sustained, unregulated IGF-1 signalling is not a low-consequence intervention.

There are no human trials for the muscle-building use. Not FDA-approved. Prohibited in sport.

Part VI — Brain, Mood and Sleep

Selank and semax

CLINICAL outside the US. Both are peptides developed in Russia and approved there — selank as an anxiolytic, semax for stroke and cognitive indications. Both are research-only in the United States.

Selank is a synthetic analogue of tuftsin studied for anxiety without sedation. Russian clinical work reported anxiolytic effects comparable to a benzodiazepine comparator, without sedation and without the dependence liability. Mechanistically it modulates GABAergic and monoaminergic signalling and increases BDNF expression.

Semax is an ACTH(4-10) analogue studied for cognition and neuroprotection, including in acute stroke. Preclinical work demonstrates increased BDNF and NGF expression and dopaminergic modulation.

The honest frame for both. These are not fringe compounds — they have decades of clinical use in one country and genuine published research. But that research has largely not been replicated to the standards Western regulators require, and the trials are generally small and often not blinded to modern standards. Approval in Russia is meaningful evidence; it is not equivalent to FDA approval. Semax was among the peptides recommended for the 503A Bulks List by the PCAC in July 2026 [22], pending FDA decision. Selank was placed in Category 2 in 2023.

No established US doses. Semax is stimulating and overstimulation or irritability is a reported effect.

Dihexa

SPECULATIVE. An angiotensin IV analogue reported in preclinical work to be dramatically more potent than BDNF at promoting synaptogenesis, and orally active with CNS penetration.

The profile is striking on paper and there are no human studies. It acts through hepatocyte growth factor and its receptor c-Met — a signalling axis with established roles in tumour growth and metastasis. A compound that powerfully activates a growth pathway implicated in cancer, with no human safety data, warrants considerable caution. It was placed in Category 2 in 2023.

P-21 and PE-22-28

SPECULATIVE. P-21 is a CNTF-derived peptide studied for neurogenesis in rodent Alzheimer's models. (It is often incorrectly described as Cerebrolysin-derived.) PE-22-28 is a spadin analogue that blocks the TREK-1 potassium channel, investigated as a fast-acting antidepressant candidate in rodents.

Both are animal-only. Neither has human trials.

DSIP (emideltide)

PRECLINICAL, with a notable regulatory data point. Delta sleep-inducing peptide was identified in the context of sleep research in the 1970s. Human research is old, small and mixed; it never established a clear hypnotic effect.

Worth noting: when the PCAC considered seven peptides in July 2026, emideltide was the one substance the committee recommended against [22] — the only rejection in a meeting that otherwise went favourably for nominators. That is a meaningful signal about the strength of its evidence base.

Part VII — Longevity and Cellular Metabolism

The most speculative section of the guide, and the one where marketing language most outruns data. Read this chapter with the evidence ratings in front of you.

Epitalon

PRECLINICAL. A synthetic tetrapeptide based on a pineal gland extract, developed by Vladimir Khavinson's group. Associated with telomerase activation and restoration of melatonin rhythm.

What the research shows. Cell culture work reports telomere elongation. Khavinson's group has published long-term observational data reporting reduced mortality in older adults.

The critical caveat. Essentially all of this research originates from a single research lineage and has not been independently replicated by outside groups. That is a significant limitation. Independent replication is how science distinguishes real effects from artefacts of a particular lab's methods, and its absence here is the central reason epitalon remains unproven despite decades of claims.

Not FDA-approved. Recommended for the 503A Bulks List by the PCAC in July 2026 [22], pending FDA decision. No established human dose.

The Khavinson bioregulators

PRECLINICAL to SPECULATIVE. A family of short peptides — pinealon, cortagen, cartalax, cardiogen, livagen, vilon, thymogen, prostamax and others — each proposed to regulate gene expression in a specific target tissue.

The organising hypothesis is that short peptides can bind DNA and modulate transcription in tissue-specific fashion. Evidence again comes predominantly from the originating research tradition, with limited independent replication. Thymalin and thymogen have the most supporting work; most of the family has very little.

Thymalin is an approved immune preparation in Russia with decades of clinical use, which places it somewhat ahead of the rest of the family.

MOTS-c

PRECLINICAL. A mitochondrial-derived peptide encoded in mitochondrial DNA, described as an exercise mimetic.

What the research shows. MOTS-c activates AMPK, improves insulin sensitivity and reduces diet-induced obesity in mice, and improves physical capacity in aged animals [10]. Circulating MOTS-c rises with exercise in humans, which is interesting observational biology.

What's missing. Human trials of MOTS-c as a therapy. Observing that a molecule rises with exercise does not establish that administering it reproduces exercise.

Not FDA-approved. Recommended for the 503A Bulks List by the PCAC in July 2026 [22], pending FDA decision.

SS-31 (elamipretide)

CLINICAL — and instructive. A mitochondria-targeting tetrapeptide that binds cardiolipin in the inner mitochondrial membrane, stabilising the electron transport chain and reducing reactive oxygen species production.

This is the most clinically advanced mitochondrial peptide, with genuine Phase 3 trials in primary mitochondrial myopathy and Barth syndrome. It is included here specifically because of what happened: **it has missed primary endpoints in several of those trials**, despite well-characterised mechanism and encouraging early data, though some open-label extensions reported benefit.

That trajectory is the most valuable thing in this chapter. A compound with strong mechanistic rationale, real investment and proper trials can still fail to demonstrate benefit. Mechanism is a hypothesis. Trials are the test.

FOXO4-DRI

SPECULATIVE in humans. A senolytic peptide designed to disrupt the FOXO4-p53 interaction, triggering apoptosis selectively in senescent cells.

In aged mice, it restored fitness, fur density and renal function in a widely cited 2017 study [11]. There is no human data — no trials, no safety database.

Senolytics are a legitimate and active field of research. This particular compound is at the earliest possible stage of translation, and deliberately inducing apoptosis in cells is not a mechanism to approach casually without human safety data.

NAD+ and its precursors

CLINICAL for raising levels; unproven for outcomes. NAD+ is a coenzyme rather than a peptide, but it sits in the same conversation.

The evidence divides cleanly. Precursors such as nicotinamide riboside and nicotinamide mononucleotide do measurably raise NAD+ levels in humans and appear well tolerated in short-term trials — that part is established. Whether raising NAD+ improves health outcomes or slows aging in humans is not established. Trials of clinical endpoints have been small and results inconsistent.

Intravenous NAD+ infusions can cause flushing, chest tightness and nausea, generally rate-dependent.

5-Amino-1MQ, SLU-PP-332 and AICAR

SPECULATIVE. Three small molecules — not peptides — grouped with this category in marketing.

5-Amino-1MQ inhibits NNMT and reduced fat mass in obese mice without changing food intake. Mouse data only; no human trials.

SLU-PP-332 is an ERR agonist that dramatically increased running endurance and fat oxidation in mice. Mouse data only; no human trials.

AICAR is an AMPK activator, the original "exercise in a pill." Human performance evidence is thin, the effective doses are impractically large, and it is prohibited in sport.

All three are sold on the strength of rodent results. None has been tested in people.

Part VIII — Bone, Skin and Sexual Health

Teriparatide and abaloparatide

APPROVED, both. The clearest demonstration in this guide that peptide drugs can be thoroughly proven.

Teriparatide (Forteo) is recombinant PTH(1-34). Given once daily, intermittent PTH exposure paradoxically stimulates bone formation rather than resorption. In the pivotal trial in postmenopausal women with prior vertebral fracture, teriparatide reduced new vertebral fractures by approximately 65% relative to placebo [12]. Approved dose: 20 micrograms subcutaneously once daily.

A note on how evidence evolves: teriparatide originally carried a boxed warning about osteosarcoma based on rat studies. After two decades of human postmarketing surveillance found no corresponding signal, the FDA removed that warning in 2020 — a good example of long-term human data overriding an animal finding.

Abaloparatide (Tymlos) is a PTHrP analogue approved in 2017. In the ACTIVE trial, it reduced new vertebral fractures by approximately 86% relative to placebo [13]. Approved dose: 80 micrograms subcutaneously once daily, with a limit on cumulative lifetime treatment duration.

Both require medical supervision, with monitoring of calcium and appropriate patient selection.

Bremelanotide (PT-141)

APPROVED. Sold as Vyleesi, approved for acquired, generalised hypoactive sexual desire disorder in premenopausal women. A melanocortin receptor agonist that acts centrally, on desire pathways in the brain, rather than on peripheral blood flow.

What the trials showed. The RECONNECT programme, two identically designed randomised trials, demonstrated statistically significant improvements in sexual desire and in distress associated with low desire [14]. The effect sizes are real but modest — worth knowing so expectations are calibrated.

Approved dosing. 1.75 mg subcutaneously as needed, at least 45 minutes before anticipated activity, with a limit of one dose per 24 hours and no more than eight per month.

Known harms. Nausea is common — it occurred in roughly 40% of participants in trials and led some to discontinue. Transient blood pressure increases and heart rate decreases occur, so it is not recommended in uncontrolled hypertension or known cardiovascular disease. Flushing and injection-site reactions are also common.

Afamelanotide (Melanotan I)

APPROVED for a specific rare disease. Sold as Scenesse, a subcutaneous implant approved to increase pain-free light exposure in adults with erythropoietic protoporphyria, a rare inherited photosensitivity disorder.

What the trials showed. Randomised trials demonstrated significantly increased pain-free time in direct sunlight for people with this condition [15].

The scope point. This approval is for a rare disease in which sun exposure causes severe pain. It is not an approval for cosmetic tanning, and the approved product is an implant administered by a healthcare professional, not an injectable sold online.

Melanotan II — the strongest caution in this guide

PRECLINICAL, and genuinely risky. Not approved in any jurisdiction. Sold widely online for tanning and libido.

The documented concerns are substantial and specific:

- Nausea and vomiting, common and often severe
- Priapism — prolonged, painful erection, which is a urological emergency that can cause permanent damage
- Darkening, enlargement and change in appearance of existing moles
- Case reports in the dermatology literature of melanoma diagnosed in users, including in changing naevi
- Systemic effects including flushing and blood pressure changes

Multiple national regulators have issued public safety warnings about Melanotan II. It was placed in FDA Category 2 in 2023.

The reason for the emphasis: a compound that induces pigmentary changes in moles while obscuring the visual cues clinicians use to detect melanoma is a poor combination, and it is being used cosmetically, by healthy people, entirely outside medical supervision. If the goals are tanning or sexual function, both have better-studied approaches. This is the one entry in this guide where the honest summary is: the risk-to-evidence ratio does not favour use.

Kisspeptin-10, gonadorelin and oxytocin

Gonadorelin is synthetic GnRH, historically FDA-approved (Factrel, Lutrepulse) for diagnostic use and pulsatile fertility treatment. Its popular use as a testosterone-therapy adjunct to maintain endogenous function is off-label and supported by mechanism and clinical practice rather than by dedicated trials.

Kisspeptin-10 — **CLINICAL.** Kisspeptin is the upstream regulator of the reproductive axis. Human research is genuinely interesting: kisspeptin has been used to trigger oocyte maturation in IVF with a lower risk of ovarian hyperstimulation syndrome than standard triggers, and has been studied for sexual and emotional brain responses. Because it acts through the body's own pulse generator, effects tend to be self-limiting. Studied via clinician-administered infusion; there is no validated at-home protocol. Placed in Category 2 in 2023.

Oxytocin — **APPROVED for obstetric use** (Pitocin, intravenous). The intranasal use for social and emotional effects is where the interesting claims live, and it's a cautionary tale about early results. After many small positive studies, the largest and most rigorous trial — a randomised placebo-controlled trial of intranasal oxytocin in children and adolescents with autism — found no significant benefit over placebo [16]. Early enthusiasm in this literature substantially exceeded what large trials confirmed.

Thymosin alpha-1

CLINICAL, approved in many countries. Sold as Zadaxin and approved in a large number of countries as an immunomodulator for hepatitis B and C and as a vaccine adjuvant. Not FDA-approved in the United States.

It modulates T-cell function and has been studied in sepsis and, during the pandemic, in severe COVID-19, where results were mixed. It has a long real-world use history and is generally well tolerated, with injection-site reactions the main complaint. Placed in Category 2 in 2023.

Part IX — Product Quality and Real Risk

If you take one chapter from this guide, this is a strong candidate.

The supply problem

For FDA-approved drugs dispensed through a pharmacy, identity, purity, sterility and dose are assured by pharmaceutical manufacturing regulation and enforced by inspection. For research-use-only peptides bought online, none of that applies.

Independent testing of consumer peptide products has repeatedly found meaningful failure rates — products that are underdosed, overdosed, degraded, contaminated, or that do not contain the labelled compound at all. Reported failure rates in purity testing have been substantial. These are not edge cases.

The specific risks worth naming:

Identity. The vial may not contain what the label says. Related peptides are far cheaper than novel ones and visually indistinguishable as lyophilised powder.

Purity. Synthesis produces truncated and deletion sequences — peptides missing one or more amino acids. These are the impurities the FDA cited in its original safety assessment, because they can provoke immune responses.

Sterility and endotoxin. Anything injected must be sterile and free of bacterial endotoxin. A certificate showing HPLC purity says nothing about either. Endotoxin contamination causes fever and systemic inflammatory reactions.

Dose accuracy. Fill weights in unregulated production vary, and a lyophilised powder gives no visual indication of its mass.

How to read a certificate of analysis

Certificates of analysis are routinely offered as proof of quality and are routinely misread.

A meaningful COA is issued by an independent third-party laboratory, names the specific lot number matching your vial, and reports identity confirmation by mass spectrometry as well as purity by HPLC. For an injectable product it should also report sterility and endotoxin testing.

Common failures: a COA supplied by the seller rather than an independent lab; a COA for a different lot; HPLC purity only, with no mass-spec identity confirmation — which tells you the material is chemically homogeneous but not what it is; no sterility or endotoxin data on an injectable; a document that is simply unverifiable because the named laboratory cannot be found or contacted.

HPLC purity is the number most often quoted and the least complete on its own.

Handling and storage

Peptides are physically fragile. Lyophilised powder should be kept sealed, cold and protected from light — refrigerated for shorter periods, frozen for longer storage. Once reconstituted, a peptide has a limited refrigerated shelf life and degrades steadily.

Two handling points that matter mechanically: add diluent slowly down the vial wall rather than directly onto the powder, and swirl rather than shake. Agitation and foaming cause physical denaturation at the air-liquid interface. A peptide that has been heated, exposed to light or shaken hard may have lost activity regardless of what the label says.

The risks that get understated

Immunogenicity. Any injected peptide can provoke antibody formation. In the worst case, antibodies cross-react with the endogenous hormone the peptide resembles, potentially interfering with a natural system. This risk is heightened by impurities and is the specific concern the FDA raised about several of these compounds.

Injection technique and infection. Subcutaneous injection carries genuine infection risk without proper aseptic technique, appropriate supplies and safe sharps disposal.

Interactions. Peptides affecting glucose metabolism, blood pressure, or hormonal axes can interact meaningfully with prescribed medications. Anyone on medication for diabetes, blood pressure, thyroid or hormonal conditions has a concrete reason to involve their prescriber.

Stacking. Combining multiple compounds multiplies unknowns rather than adding them. If something goes wrong on a five-compound stack, you have no way to identify the cause. Combination products sold as blends have generally not been studied as combinations — their evidence, where it exists, belongs to the individual ingredients studied separately.

Part X — Questions Worth Asking

A practical checklist, applicable to any compound in this guide or any you encounter later.

About the evidence

1. Has this been tested in humans, for the use I care about? Not "is there research" — is there *human* research, on *this* application?
2. If the studies are in animals, what species, what dose, and how does that dose scale to a human?
3. Who conducted and funded the research, and has any independent group replicated it?
4. Are there negative trials? A compound with only positive results and no failures is a warning sign, not a strong record.

About regulatory status

1. Is it FDA-approved for anything? If yes, is my intended use the approved one?
2. If it's approved abroad, what was that regulator's evidence standard?
3. What is its current 503A status — and is the information I'm reading current, given how much moved in 2026?
4. Is it prohibited in sport, and does that affect me?

About the product

1. Is there a third-party COA matching this specific lot?
2. Does it confirm identity by mass spectrometry, not just purity by HPLC?
3. For an injectable: is there sterility and endotoxin testing?
4. What happens if the product is wrong — is there recourse?

About your own situation

1. Does a clinician who knows my history know I'm considering this?
2. How does it interact with my current medications?
3. What specific outcome am I expecting, and how will I know whether it happened?
4. What would make me stop?

Question 15 deserves emphasis. Without a defined expected outcome and a way to measure it, it becomes very difficult to tell whether a compound did anything at all — and easy to attribute normal variation, placebo response, or the effects of simultaneous changes in diet and training to the compound.

Quick Reference

Evidence ratings reflect the state of published human evidence for the commonly promoted use, not commercial popularity.

Compound	Category	Rating	Status
Semaglutide	Metabolic	Approved	FDA approved (Wegovy, Ozempic)
Tirzepatide	Metabolic	Approved	FDA approved (Zepbound, Mounjaro)
Orforglipron	Metabolic	Approved	FDA approved 2026 (not a peptide)
Retatrutide	Metabolic	Clinical	Phase 3; not approved
CagriSema	Metabolic	Clinical	NDA under FDA review
Survodutide	Metabolic	Clinical	Phase 3; not approved
Mazdutide	Metabolic	Clinical	Approved in China only
AOD-9604 / HGH Frag	Metabolic	Preclinical	Failed Phase 2b for obesity
BPC-157	Recovery	Preclinical	Not approved; 503A pending
TB-500	Recovery	Preclinical	Not approved; 503A pending
GHK-Cu (topical)	Skin	Clinical	Cosmetic ingredient; Cat. 1 non-injectable
GHK-Cu (injected)	Recovery	Preclinical	Category 2 (2023)
KPV	Recovery	Preclinical	Not approved; 503A pending
ARA-290	Nerve	Clinical	Small positive RCT; not approved
LL-37	Immune	Preclinical	Two-sided risk profile
Tesamorelin	GH	Approved	FDA approved (Egrifta)
Sermorelin	GH	Historical	Withdrawn; compounded
CJC-1295	GH	Preclinical	Not approved; banned in sport
Ipamorelin	GH	Preclinical	Not approved; banned in sport
GHRP-2 / -6, hexarelin	GH	Preclinical	Not approved; banned in sport
IGF-1 LR3 / DES	Growth	Preclinical	Proliferation and hypoglycaemia risk
Selank	Brain	Clinical (RU)	Approved in Russia only
Semax	Brain	Clinical (RU)	Approved in Russia; 503A pending

Compound	Category	Rating	Status
Dihexa	Brain	Speculative	No human data; c-Met concern
P-21, PE-22-28	Brain	Speculative	Animal only
DSIP (emideltide)	Sleep	Preclinical	PCAC voted against, 2026
Epitalon	Longevity	Preclinical	Single research lineage; 503A pending
Bioregulators	Longevity	Preclinical	Limited independent replication
MOTS-c	Cellular	Preclinical	Not approved; 503A pending
SS-31 (elamipretide)	Cellular	Clinical	Missed primary endpoints
FOXO4-DRI	Cellular	Speculative	Animal only
NAD+ precursors	Cellular	Clinical	Raise levels; outcomes unproven
5-Amino-1MQ, SLU-PP-332	Cellular	Speculative	Mouse only; not peptides
Teriparatide	Bone	Approved	FDA approved (Forteo)
Abaloparatide	Bone	Approved	FDA approved (Tymlos)
Bremelanotide (PT-141)	Sexual	Approved	FDA approved (Vyleesi)
Afamelanotide	Skin	Approved	FDA approved (Scenesse), rare disease
Melanotan II	Skin	Preclinical	Not approved; regulator warnings
Kisspeptin-10	Reproductive	Clinical	Clinician-administered research
Gonadorelin	Reproductive	Approved	Approved; popular use is off-label
Oxytocin (nasal)	Social	Clinical	Largest RCT was negative
Thymosin alpha-1	Immune	Clinical	Approved in many countries, not US

Glossary

Agonist — A molecule that binds a receptor and activates it. An antagonist binds and blocks it.

Amino acids — The building blocks that link together to form peptides and proteins.

Bioavailability — The proportion of an administered dose that reaches systemic circulation in active form.

503A Bulks List — The FDA list of bulk drug substances that may be used by compounding pharmacies. Category 1 substances may be used pending final rulemaking; Category 2 substances raise significant safety concerns [21].

cGMP — Current Good Manufacturing Practice, the regulatory standard for pharmaceutical manufacturing. Not applicable to research-use-only products.

COA (Certificate of Analysis) — A laboratory document reporting testing results for a specific production lot.

Endotoxin — Bacterial cell wall components that cause fever and inflammation if injected, even when no live bacteria are present.

GLP-1 — Glucagon-like peptide-1, a gut incretin hormone that signals satiety and stimulates insulin release.

Half-life — The time for the concentration of a compound in the body to fall by half.

HPLC — High-performance liquid chromatography, used to assess purity. It does not confirm identity.

IGF-1 — Insulin-like growth factor-1, the principal mediator of growth hormone's effects.

Immunogenicity — The capacity of a substance to provoke an immune response against itself.

Lyophilised — Freeze-dried. The stable powder form most peptides are supplied in.

Mass spectrometry — An analytical method that confirms molecular identity by measuring mass-to-charge ratio.

Off-label — Use of an approved drug for a purpose, population or dose other than the approved one.

Preclinical — Research conducted before human trials, in cells and animals.

Reconstitution — Dissolving a lyophilised peptide in diluent, typically bacteriostatic water.

Secretagogue — A substance that causes another substance to be secreted. GH secretagogues prompt endogenous growth hormone release.

Senolytic — A compound designed to selectively eliminate senescent cells.

Subcutaneous — Into the tissue layer beneath the skin.

WADA — The World Anti-Doping Agency. Its S0 category covers substances with no health authority approval, which captures most research peptides.

References

Peer-reviewed sources are cited with DOI and PubMed ID. Regulatory sources are cited to the issuing agency's own published document wherever one exists, in preference to secondary commentary. Where this guide describes a compound as lacking human evidence, that absence is itself the finding — there is no citation to give, and any source claiming otherwise should be checked against the primary literature.

Clinical trials and peer-reviewed literature

- [1] Wilding JPH, Batterham RL, Calanna S, et al. Once-Weekly Semaglutide in Adults with Overweight or Obesity (STEP 1). *New England Journal of Medicine*. 2021;384(11):989–1002. doi:10.1056/NEJMoa2032183. PMID 33567185. NCT03548935.
- [2] Lincoff AM, Brown-Frandsen K, Colhoun HM, et al. Semaglutide and Cardiovascular Outcomes in Obesity without Diabetes (SELECT). *New England Journal of Medicine*. 2023;389(24):2221–2232. doi:10.1056/NEJMoa2307563. PMID 37952131. NCT03574597.
- [3] Jastreboff AM, Aronne LJ, Ahmad NN, et al. Tirzepatide Once Weekly for the Treatment of Obesity (SURMOUNT-1). *New England Journal of Medicine*. 2022;387(3):205–216. doi:10.1056/NEJMoa2206038. PMID 35658024. NCT04184622.
- [4] Jastreboff AM, Kaplan LM, Frías JP, et al. Triple-Hormone-Receptor Agonist Retatrutide for Obesity — A Phase 2 Trial. *New England Journal of Medicine*. 2023;389(6):514–526. doi:10.1056/NEJMoa2301972. PMID 37366315. NCT04881760.
- [5] Wilding JPH, Batterham RL, Davies M, et al. Weight regain and cardiometabolic effects after withdrawal of semaglutide: the STEP 1 trial extension. *Diabetes, Obesity and Metabolism*. 2022;24(8):1553–1564. doi:10.1111/dom.14725. PMID 35441470.
- [6] Pickart L, Margolina A. Regenerative and Protective Actions of the GHK-Cu Peptide in the Light of the New Gene Data. *International Journal of Molecular Sciences*. 2018;19(7):1987. doi:10.3390/ijms19071987.
- [7] Falutz J, Allas S, Blot K, et al. Metabolic Effects of a Growth Hormone-Releasing Factor in Patients with HIV. *New England Journal of Medicine*. 2007;357(23):2359–2370. doi:10.1056/NEJMoa072375.
- [8] Stanley TL, Feldpausch MN, Oh J, et al. Effect of Tesamorelin on Visceral Fat and Liver Fat in HIV-Infected Patients with Abdominal Fat Accumulation. *JAMA*. 2014;312(4):380–389. doi:10.1001/jama.2014.8334.
- [9] Teichman SL, Neale A, Lawrence B, et al. Prolonged stimulation of growth hormone and insulin-like growth factor I secretion by CJC-1295, a long-acting analog of GH-releasing hormone, in healthy adults. *Journal of Clinical Endocrinology & Metabolism*. 2006;91(3):799–805. doi:10.1210/jc.2005-1536.
- [10] Lee C, Zeng J, Drew BG, et al. The Mitochondrial-Derived Peptide MOTS-c Promotes Metabolic Homeostasis and Reduces Obesity and Insulin Resistance. *Cell Metabolism*. 2015;21(3):443–454. doi:10.1016/j.cmet.2015.02.009.

- [11] Baar MP, Brandt RMC, Putavet DA, et al. Targeted Apoptosis of Senescent Cells Restores Tissue Homeostasis in Response to Chemotoxicity and Aging. *Cell*. 2017;169(1):132–147.e16. doi:10.1016/j.cell.2017.02.031.
- [12] Neer RM, Arnaud CD, Zanchetta JR, et al. Effect of Parathyroid Hormone (1-34) on Fractures and Bone Mineral Density in Postmenopausal Women with Osteoporosis. *New England Journal of Medicine*. 2001;344(19):1434–1441. doi:10.1056/NEJM200105103441904.
- [13] Miller PD, Hattersley G, Riis BJ, et al. Effect of Abaloparatide vs Placebo on New Vertebral Fractures in Postmenopausal Women with Osteoporosis (ACTIVE). *JAMA*. 2016;316(7):722–733. doi:10.1001/jama.2016.11136.
- [14] Kingsberg SA, Clayton AH, Portman D, et al. Bremelanotide for the Treatment of Hypoactive Sexual Desire Disorder: Two Randomized Phase 3 Trials (RECONNECT). *Obstetrics & Gynecology*. 2019;134(5):899–908. doi:10.1097/AOG.0000000000003500.
- [15] Langendonk JG, Balwani M, Anderson KE, et al. Afamelanotide for Erythropoietic Protoporphyria. *New England Journal of Medicine*. 2015;373(1):48–59. doi:10.1056/NEJMoa1411481.
- [16] Sikich L, Kolevzon A, King BH, et al. Intranasal Oxytocin in Children and Adolescents with Autism Spectrum Disorder. *New England Journal of Medicine*. 2021;385(16):1462–1473. doi:10.1056/NEJMoa2103583.
- [17] Vasireddi N, Hahamyan H, Salata MJ, Karns M, Calcei JG, Voos JE, Apostolakos JM. Emerging Use of BPC-157 in Orthopaedic Sports Medicine: A Systematic Review. *HSS Journal*. 2025;21(4):485–495. doi:10.1177/15563316251355551. PMID 40756949. — *The systematic review establishing that of 544 articles screened (1993–2024), 36 met inclusion criteria, of which 35 were preclinical and one was a small uncontrolled clinical case series.*
- [18] McGuire FP, Martinez R, Lenz A, Skinner L, Cushman DM. Regeneration or Risk? A Narrative Review of BPC-157 for Musculoskeletal Healing. *Current Reviews in Musculoskeletal Medicine*. 2025;18(12):611–619. doi:10.1007/s12178-025-09990-7. PMID 40789979.
- [19] Sanyal AJ, Kaplan LM, Frias JP, et al. Triple hormone receptor agonist retatrutide for metabolic dysfunction-associated steatotic liver disease: a randomized phase 2a trial. *Nature Medicine*. 2024. PMID 38858523.

Regulatory sources (primary)

- [20] U.S. Food and Drug Administration. *Certain Bulk Drug Substances for Use in Compounding that May Present Significant Safety Risks*. Center for Drug Evaluation and Research. Content current as of April 22, 2026. — *The source of all FDA safety-risk language quoted in this guide, including the entries for BPC-157, TB-500, KPV, CJC-1295, ipamorelin, GHRP-2, GHRP-6, ibutamoren, Melanotan II, epitalon, MOTS-c, dihexa, emideltide, semax, selank, LL-37, AOD-9604 and injectable GHK-Cu.* <https://www.fda.gov/drugs/human-drug-compounding/certain-bulk-drug-substances-use-compounding-may-present-significant-safety-risks>
- [21] U.S. Food and Drug Administration. *Bulk Drug Substances Used in Compounding Under Section 503A of the FD&C Act* — the interim policy and Category 1/2/3 framework. <https://www.fda.gov/drugs/human-drug-compounding/bulk-drug-substances-used-compounding-under-section-503a-fdc-act>

[22] U.S. Food and Drug Administration. *July 23–24, 2026: Meeting of the Pharmacy Compounding Advisory Committee*. Docket No. FDA-2025-N-6895. <https://www.fda.gov/advisory-committees/advisory-committee-calendar/july-23-24-2026-meeting-pharmacy-compounding-advisory-committee-07232026>

[23] U.S. Food and Drug Administration. *Briefing Document, Pharmacy Compounding Advisory Committee Meeting, July 2026*. — *The document in which FDA proposed that none of the fourteen peptide forms under consideration be included on the 503A Bulks List*. <https://www.fda.gov/media/193342/download>

[24] U.S. Food and Drug Administration. *Questions for the Pharmacy Compounding Advisory Committee, July 23–24, 2026* — the voting questions as put to the committee. <https://www.fda.gov/media/193711/download>

[25] U.S. Food and Drug Administration. *Understanding the Risks of Compounded Drugs*. <https://www.fda.gov/drugs/human-drug-compounding/understanding-risks-compounded-drugs>

[26] World Anti-Doping Agency. *The 2026 Prohibited List — World Anti-Doping Code International Standard*, effective January 1, 2026. BPC-157 is named explicitly within class S0 (Non-Approved Substances); growth hormone secretagogues and growth factors fall under class S2. <https://www.wada-ama.org/en/prohibited-list>

Trial announcements and regulatory filings

[27] Eli Lilly and Company. *Retatrutide delivered weight loss of up to an average of 71.2 lbs along with substantial relief from osteoarthritis pain in first successful Phase 3 trial (TRIUMPH-4)*, December 11, 2025. NC-T05931367. <https://investor.lilly.com/news-releases>

[28] Eli Lilly and Company. *Retatrutide delivered powerful weight loss in pivotal Phase 3 obesity trial (TRIUMPH-1)*, May 21, 2026. NCT05929066.

[29] Novo Nordisk. REDEFINE 1 — Phase 3 results for cagrilintide-semaglutide (CagriSema) in adults with obesity or overweight, 2025; and REDEFINE 2 in adults with type 2 diabetes, 2025.

[30] Novo Nordisk. New Drug Application submitted to the FDA for CagriSema, December 18, 2025.

[31] U.S. Food and Drug Administration approval of orforglipron (Foundayo), April 2026 — the first oral non-peptide GLP-1 receptor agonist; ATTAIN-1 Phase 3 programme.

A note on secondary commentary. Legal and consultancy analyses of the 2023 Category 2 additions, the April 2026 removals, and the July 2026 PCAC votes were consulted in preparing this guide and are useful for practitioners, but every factual claim about FDA action in this guide is sourced to the agency's own documents listed above. Readers are encouraged to check those primary documents directly, since this area continues to move.

Disclaimer

This guide is provided for **educational and informational purposes only**. It is not medical advice, it does not establish a clinician-patient relationship, and it is not a substitute for consultation with a qualified, licensed healthcare professional. Nothing in it is a recommendation, instruction or encouragement to obtain, possess or use any compound.

On approval status. Many compounds described here are not approved by the FDA or any comparable regulator for human use. Products sold as "research use only" are not authorised for human consumption, are not manufactured under pharmaceutical quality standards, and are not subject to premarket review for safety, efficacy, identity, purity or sterility. Where a compound is an approved medicine, that approval applies only to the specific indication, population and dose described in its labelling, and requires a prescription and medical supervision.

On dosing. This guide deliberately does not provide dosing information for compounds that are not FDA-approved, because no authoritative human dosing data exists for them. Doses stated for approved medicines are reproduced from approved labelling for reference and are not a recommendation for any individual.

On risk. Peptides can cause adverse effects, can interact with prescribed medications, and — when obtained outside the regulated supply chain — may be impure, contaminated, degraded, mislabelled or counterfeit. Several compounds described here are prohibited in competitive sport.

On currency. The regulatory landscape described in this guide changed substantially during 2023 and again in 2026 and continues to move. Statements about legal and regulatory status were accurate to the best of our knowledge at the time of writing and should be independently verified before being relied upon. Scientific findings described here reflect published evidence at the time of writing and may be superseded.

Please speak with a knowledgeable, licensed healthcare professional before considering any compound described in this guide.

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